

IIoT Secure Gateway

Secure Prototype Design with FPGA based Data Diode

Objectives

- Safeguard OT network and PLCs from IT-side threats
- Implement strict unidirectional data flow from OT to IT
- Create a digital twin for safe IT monitoring and analysis
- Design prototype with minimal cost and complexity
- Enable future extensibility with real OT hardware

System Architecture Overview

Virtual OT

OpenPLC simulators and protocol emulators generating realistic OT data

FPGA Based Diode

One-way data forwarding with optical media separation

IT Network

Safe access to digital twin for monitoring and analytics

Virtual OT Environment

- OpenPLC: Open-source PLC simulator for device emulation
- ModbusPal: Modbus TCP/RTU protocol simulator
- OpenSCADA: Modular SCADA system with simulation
- Tcpreplay: Replay captured OT network traffic
- Scapy: Custom OT protocol packet generation

FPGA Data Diode: Physical Layer

- Optical fiber transmitter on OT side
- Optical receiver on IT side (unidirectional)
- Prevents reverse-path data or control signals
- Physical enforcement independent of software logic
- Supports tamper detection and alarms

FPGA Design (Phase 1)

- Receiver Block: Clock recovery, frame decoding, CRC validation
- Packet Buffer: Block RAM for speed matching and jitter handling
- Forwarding Engine: Unidirectional packet streaming logic
- Transmitter Block: Encoding and IT-side media driving

FPGA Design (Phase 2)

- Management Processor: Softcore CPU for configuration and monitoring
- Tamper Detection: Hardware monitoring of physical layer integrity
- Status Logging: Event capture and alerting on anomalies
- Initial Logic: Simple one-way forwarding, extensible for filtering

Hardware Components and Cost

FPGA Board	INR 25,000 - 60,000
Optical Transceivers	INR 2,000 - 5,000
Switches and Cables	INR 2,000 - 5,000
Fiber Cable	INR 3000
Server	
Total	INR 32,000 - 70,000

Network Infrastructure

- Gigabit Ethernet switch (managed for VLAN support)
- VLAN-based logical separation of OT and IT networks
- Static IP addressing for test stability
- No routing between OT and IT outside FPGA diode
- Future extensibility: passive taps for real OT integration

Digital Twin Server Role

Single server hosts both virtual OT simulation and digital twin functions:

- Runs PLC simulators generating OT protocol traffic
- Receives replicated data via FPGA for state reconstruction
- Provides safe access point for IT monitoring and analytics
- Docker containers for modular simulator deployment

IT Test System Access

- Connect only to digital twin environment (no direct OT access)
- Run monitoring, analytics, and cybersecurity tools
- Access replicated OT state safely for visualization
- Test intrusion detection and anomaly analysis
- Perform proactive threat hunting on OT data

Implementation Phases

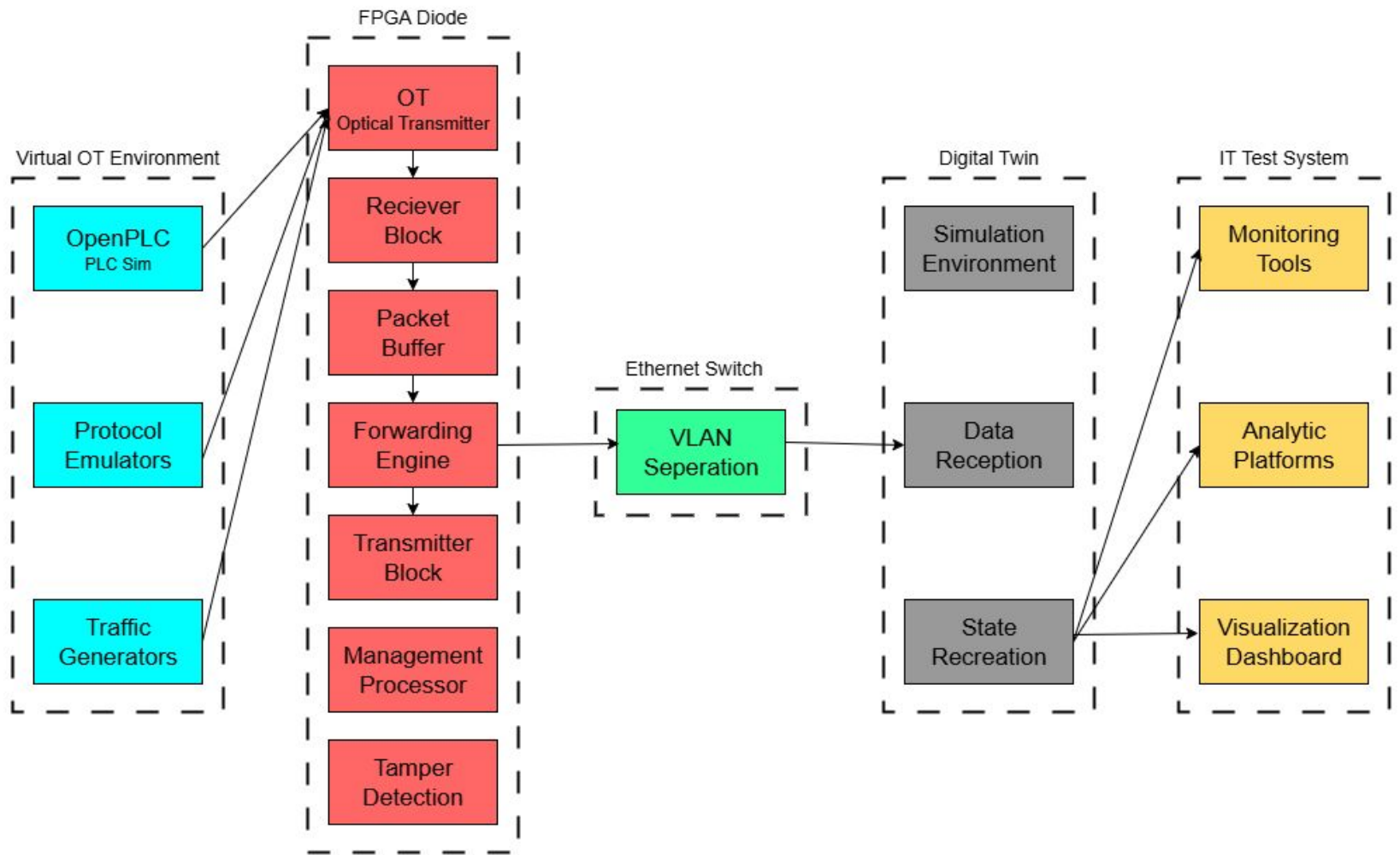
- **Phase 1:** Set up virtual OT with OpenPLC simulators
- **Phase 2:** Implement basic FPGA one-way forwarding logic
- **Phase 3:** Deploy digital twin server with monitoring
- **Phase 4:** Add protocol filtering and security checks

Key Advantages

- Physical one-way separation at hardware level ensures OT safety
- Digital twin enables IT visibility without OT exposure
- Low-cost prototype suitable for experimentation
- Modular design allows incremental feature addition

Future Enhancements

- Deep packet inspection for protocol-level security
- Real OT network integration via passive taps
- Advanced threat detection and anomaly analysis
- Predictive maintenance analytics on twin





Thank You